

Springer Texts in Statistics

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An Introduction to Statistical Learning

with Applications in R

 Springer

ANALYTICS

Week 5

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*Examples Inspired By
CAS, SOA, Coaching
Actuaries ©*

Key Formulas This Week

- $se(b_j) = s\sqrt{(j+1)st \text{ diagonal of } (\mathbf{X}'\mathbf{X})^{-1}}$
- $Cov(b_i, b_j) = s^2[(i+1, j+1)st \text{ element of } (\mathbf{X}'\mathbf{X})^{-1}]$
- $t(b_j) = \frac{b_j - H_0(b_j)}{se(b_j)} \text{ w/df} = n - (p+1)$
- $r_{\text{partial}} = \frac{t_i}{\sqrt{t_i^2 + n - (p+1)}}$
- CI/PI: Estimate $\pm t\text{-Critical Value} * \text{Standard Error}$
- $F = \frac{\frac{SSReg}{p}}{\frac{SSE}{n-(p+1)}} = \frac{\frac{R^2}{1-R^2}}{\frac{p}{n-(p+1)}} \text{ or } F = \frac{\frac{(SSE_0 - SSE_{full})}{q}}{\frac{SSE_{full}}{n-(p+1)}} = \frac{\frac{R_{full}^2 - R_{reduced}^2}{q}}{\frac{1-R_{full}^2}{n-(p+1)}}$

Coefficient Interpretation

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-1.3863	0.1685	-8.225	< 2e-16	***
ALow	0.7133	0.2980	2.394	0.0167	*
BLow	-1.3940	0.2115	-6.590	4.41e-11	***
ALow:BLow	1.4142	0.3467	4.079	4.53e-05	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

- ANormal, BNormal are the base categories.

Parameter SEs

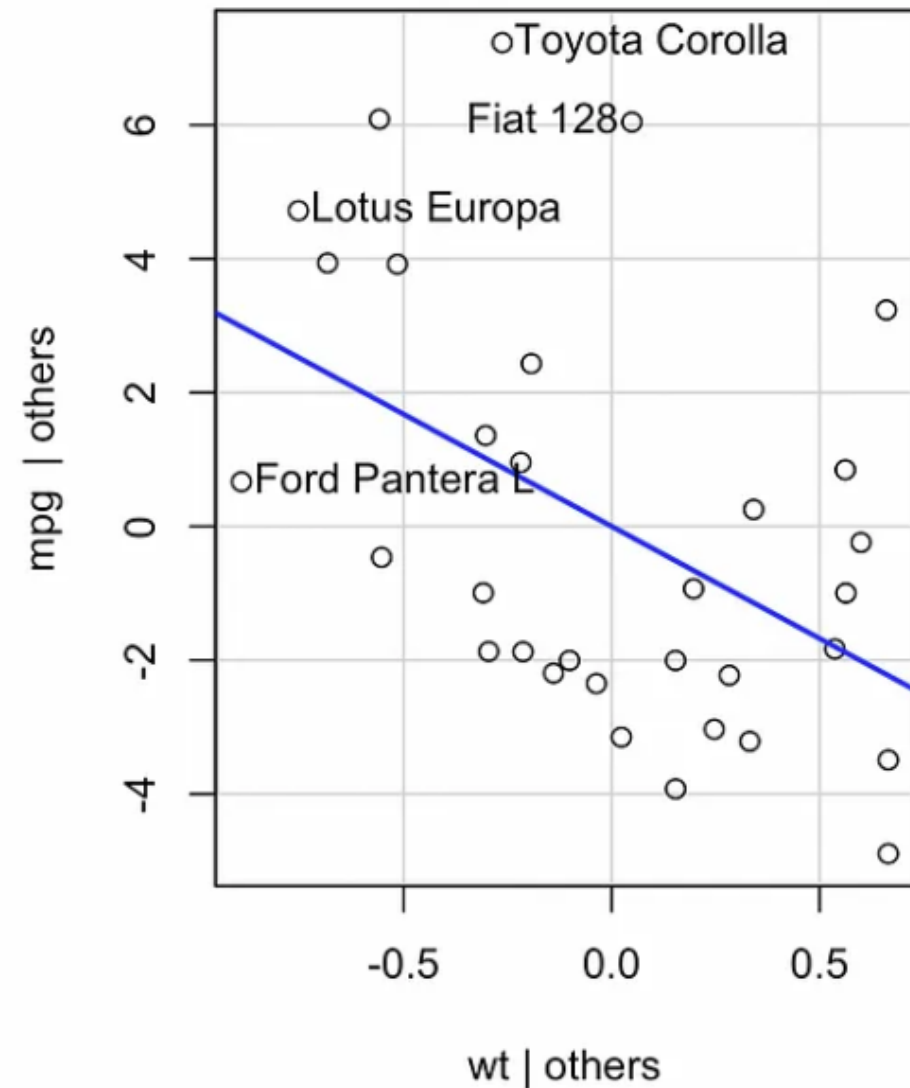
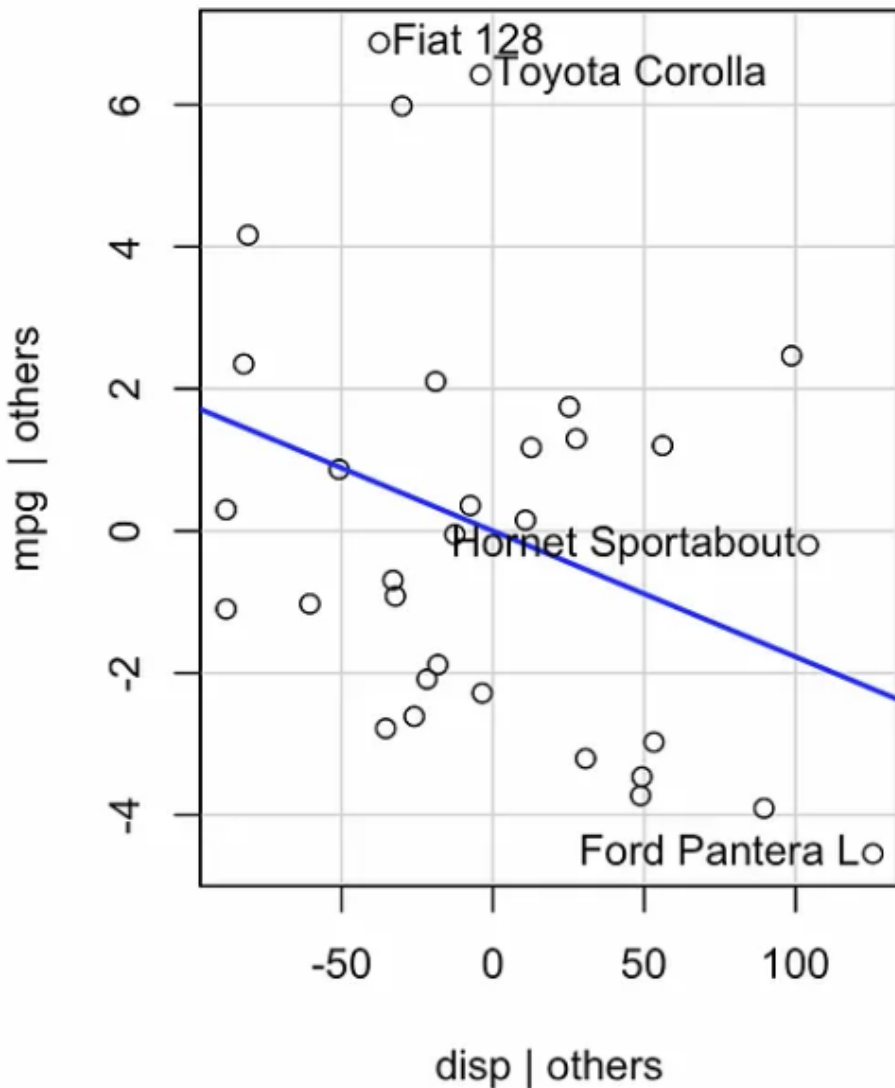
- Standard Error =
- $se(b_j) = s\sqrt{(j + 1)\text{st diagonal of } (\mathbf{X}'\mathbf{X})^{-1}}$
- $Cov(b_i, b_j) = s^2[(i + 1, j + 1)\text{st element of } (\mathbf{X}'\mathbf{X})^{-1}]$

Added Variable Plots

- **Added variable plots** (partial regression) try to
 - 1. Run a regression $y \sim x_1 + x_2 + \cdots x_p$ and store the residuals e_y .
 - 2. Run a regression $x_{p+1} \sim x_1 + x_2 + \cdots x_p$ and store the residuals e_x .
 - 3. Plot e_y against e_x .

Added Variable Plot

Added-Variable Plots



Partial Correlation

- The **partial correlation coefficient** is the

- $r_{partial} =$

- p : Total predictors including the i^{th}

- $t_i = \frac{b_i}{se(b_i)}$

Predictors Needed *

- The # of explanatory predictors needed quantifies model flexibility.
- Each continuous explanatory variable adds 1 predictor.
- Each categorical explanatory variable adds $m - 1$ predictors.
- Continuous w/continuous interaction term adds 1 predictor.
- Continuous w/categorical interaction term adds $m - 1$ predictors.
- n -level categorical w/ m -level categorical interaction term adds $(n - 1)(m - 1)$ predictors

1. # Predictors Illustration

- You are given a multiple linear regression model of claim size with the following potential explanatory variables only:
- House value, a continuous variable modeled with a 4th degree polynomial
- Owner age, a continuous variable modeled linearly
- Owner Gender, a categorical variable with 2 levels
- # of occupants, a categorical variable with 5 levels
- One interaction term, between gender and # of occupants.
- Determine the max # of parameters in this model.

t -Test

- Statistical hypothesis tests help determine whether explanatory variables x_j are important by testing $H_0: \beta_j = 0$ vs. $H_a: \beta_j \neq 0$.
- Establish a significance level α .
- $t(b_j) = \frac{b_j - H_0(b_j)}{se(b_j)}$, $df = n - (p + 1)$ if assumptions 1-5 hold
- Reject H_0 if $|t(b_j)| > t_{n-(p+1), \frac{\alpha}{2}}$ for 2-sided
- Or, just use the p -value output directly.

t -Test Summary *

- Small $p \rightarrow$ Higher magnitude t -statistic, more evidence to reject $H_0 \rightarrow$ Keep parameter in model (it's significant)
- Large $p \rightarrow$ Smaller magnitude t -statistic, fail to reject $H_0 \rightarrow$ Drop parameter from model (confidence interval overlaps 0)
- The pre-set standard for rejecting H_0 is α .

Example 1

- You wish to explain y (Math score) using a MLR model w/16 observations:
- $\hat{y} = b_0 + b_1x_1 + b_2x_2 + b_3x_3 + b_4x_4$
- Sleep has 3 levels: Low, Medium (Base), High
- For each explanatory variable, you decide to test the null hypothesis $H_0: b_i = 0$ at $\alpha = 10\%$ significance.
- A. Which variable can be removed without data adjustment?
- B. What is the partial correlation coefficient between Math score and Hours?

	Coefficient	Standard Error
$\beta_1 = \text{SleepLow}$	-3	1
$\beta_2 = \text{SleepHigh}$	5	5
$\beta_3 = \text{Hours}$	8	3.5
$\beta_4 = \# \text{ Coffees}$	9	6

Example 1 Sol

- $t(b_j) = \frac{b_j}{se(b_j)}, df = n - (p + 1)$

- $r_{partial} = \frac{t_i}{\sqrt{t_i^2 + n - (p + 1)}}$

Confidence Intervals

- Confidence intervals provide a range of reliability for each parameter estimate, given a chosen significance level α .
- CI: $b_j \pm t_{critical} * se(b_j)$
- Estimate $\pm$

Prediction Intervals

- Prediction intervals provide a range of reliability for each $\hat{y}_{pred}$ from a model, and are wider due to
- FYI PI: $\hat{y}_{pred} \pm t_{critical} \sqrt{s^2 (1 + \mathbf{x}_*^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_*^T)}$
 - $\mathbf{x}_*^T$: Vector of new inputs for prediction

Prediction Intervals 2

- Prediction accuracy is limited by:
- Reducible error
- Model bias
- Irreducible error
- Use prediction intervals for individual predictions, confidence intervals for means, parameters estimates.

2. CI Illustration

- Consider the following dataset. You fit a regression model to y using two explanatory variables, x_1 and x_2 .
- From the fitted model, root mean squared error is 1.373.
- $b = \begin{pmatrix} 0.15 \\ 0.69 \\ 2.88 \end{pmatrix}, (X'X)^{-1} = \begin{pmatrix} 0.54 & -0.077 & -0.15 \\ -0.077 & 0.154 & -0.69 \\ -0.15 & -0.69 & 4.12 \end{pmatrix}$
- Determine the 95% confidence interval for b_2 .

i	1	2	3	4
$x_{i,1}$	-1	2	4	6
$x_{i,2}$	0	0	1	1
y_i	0	1	5	8

2. CI Illustration 2

- CI: $b_j \pm t_{critical} * se(b_j)$
- $se(b_j) = s\sqrt{(j+1)st \text{ diagonal of } (X'X)^{-1}}$

Example 2

- A multiple linear regression model built on 3 explanatory variables and 18 observations has a 95% confidence interval for b_2 of $(-3.6, -1)$.
- Find the t -statistic for testing $H_0: b_2 = 0$.
- Is the explanatory variable associated with b_2 significant? Should it be kept or removed from the model?

Example 2 Sol

- CI: $b_j \pm t_{critical} * se(b_j)$
- $t(b_j) = \frac{b_j}{se(b_j)}, df = n - (p + 1)$

ANOVA

- The **ANOVA** (analysis of variance) table summarizes the sources of variability.
- Mean square is the sum of square values divided by its respective degrees of freedom.

Source	Sum of Squares	df	Mean Square
Regression	$SSReg$	p	$MSR = \frac{SSReg}{p}$
Error	SSE	$n - (p + 1)$	$MSE = \frac{SSE}{n - (p + 1)}$
Total	SST	$n - 1$	

2 *F*-test Versions

- 1. Determine significance of an entire model or
- 2. Test whether a more complex model is better than a simpler nested version
- A predictor can be significant by a t-test while a whole model F-test is not significant because they test different hypotheses.

Whole F -test

- To test an entire model, $H_0: \beta_1 = \beta_2 = \cdots \beta_p = 0$
- H_a :
- $F_{p, n-(p+1)} = \frac{MSReg}{MSE} =$
 - p : # of explanatory variables (excludes fitted intercept)
 - R^2 : of full model

SLR F -test

- For basic linear regression, the F -statistic is directly related to the t -statistic:
- $t =$
- Higher $MSReg$ relative to MSE is more evidence to

Nested F -test

- To compare whether a subset model is better,
- H_0 (reduced): A subset q of the β_i coefficients are 0
- H_a (full): At least one of the β_i in the subset is non-zero

- $$F = \frac{\frac{(SSE_0 - SSE_{alt})}{q}}{\frac{SSE_{alt}}{n - (p + 1)}} =$$

- $t = \sqrt{F}$ if 1 predictor is omitted for H_0 vs. H_a .

3. F -test Illustration (SOA)

- A. For a simple linear regression model with 8 observations, if $\sum_{i=1}^8 (y_i - \hat{y}_i)^2 = 200$ and the F statistic is 3, calculate the coefficient of determination.
- B. For the multiple regression model $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \beta_5 x_5 + \epsilon$ to estimate house size, you are given:
 - The # of observations is 3,120.
 - The total sum of squares is 15,000.
 - $H_0: \beta_3 = \beta_4 = \beta_5 = 0$
 - The coefficient of determination for the full model is 40%.
 - The regression sum of squares for the reduced model is 5,500.
 - Determine the value of the F -statistic for testing H_0 .

3. F -test Illustration

- SLR: $F = \frac{MSReg}{MSE} = (n - 2) \frac{R^2}{1 - R^2}$
- $R^2 = \frac{SSR}{SST} = \frac{SST - SSE}{SST}$
- $F = \frac{\frac{(SSE_0 - SSE_{alt})}{q}}{\frac{SSE_{alt}}{n - (p + 1)}} = \frac{\frac{R_{alt}^2 - R_0^2}{q}}{\frac{1 - R_{alt}^2}{n - (p + 1)}}$

Example 3 (SOA)

- Twenty pairs of observations (x_i, y_i) are fitted to a linear regression model A: $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$.
- $\sum x_i = 100$
- $\sum y_i = 200$
- $\sum x_i^2 = 2,100$
- $\sum y_i^2 = 8,000$
- $b_1 = 1.1$
- The F -ratio for testing $H_0: \beta_1 = 0$ in model A equals the F -ratio for testing $H_0: \beta_1 = \beta_2 = 0$ in model B on the same data with an extra predictor x_{2i} : $y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \varepsilon_i$.
- What must be the regression sum of squares for Model B?

Example 3 Sol

- $F_{p, n-(p+1)} = \frac{MSReg}{MSE} = \frac{\frac{SSReg}{p}}{\frac{SSE}{n-(p+1)}} = \frac{R^2}{1-R^2} * \frac{n-(p+1)}{p}$
- $SST = \sum (y_i - \bar{y})^2 = \sum y_i^2 - \frac{(\sum y_i)^2}{n}$
- $SSReg = \sum (\hat{y}_i - \bar{y})^2 = b_1^2 \left(\sum x_i^2 - \frac{(\sum x_i)^2}{n} \right)$ in SLR

Summary Check

- *Can you describe in your own words:*
- How an added variable plot works and the partial correlation formula
- How to count number of predictors
- How to conduct a t -test
- How a full-model F -test vs. partial F -test work and their formulas
- How to find confidence and prediction intervals